Cleaning and Merging Datasets

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Methods

Genetic and phenotypic data collected by the team of Dr George Perry at Pennsylvania State University with the University of Antananarivo. Dr Jason Hodgson and Dr Rindra Rakotoarivony led this data collection in 2015. These data comprise of 192 individuals from five regions: Merina people from the Central Highlands, Betsileo people from the southern Highlands, Sakalava and Tsimihety people from the northwestern coast, Betsimisaraka people from the eastern coast, and people of various ethnic groups from the extreme north in the town of Diego. Samples were collected from married couples, and ~690,000 SNPs were typed using the Illumina HumanOmniExpress-24 array.

Two main datasets were created, a Malagasy only dataset and a global dataset combining Malagasy samples with comparative data. The Malagasy dataset was created by merging the data from this study with Malagasy data from (Pierron et al. 2014) (n = 69). For a comparative global dataset, this data was combined with data from southeast Asian and African populations in the (Auton et al. 2015) dataset (n = 2479), individuals from Borneo from (Brucato et al. 2016) (n = 39), and individuals from the islands of Sumba (n = 30, subsampled at random from 240) and Timor (n = 7) from (Cox et al. 2016).

Scripts in R were used to map data to GRCH38 from build GRCH37. The package BiomaRt (version '2.56.1'), which utilises the ensembl database, was employed to do so (Durinck et al. 2005; Durinck et al. 2009). Data were then cleaned and merged in R with package tidypopgen (Carter and Manica 2025). Only autosomal SNPs were analysed. All SNPs with a missingness rate above 0.01 across all samples were removed. All individuals with a genotype call rate less than 0.9 (missingness exceeding 0.1) were removed. All SNPs with minor allele frequency below a threshold of 0.025 were removed. All SNPs with a Hardy-Weinberg equilibrium test p-value below 0.01 were removed. Close relatives were removed using the KING robust kinship coefficient (Manichaikul et al. 2010), implemented in tidypopgen. One individual was removed from each pair with a kinship coefficient greater than 0.0884, equivalent to removing relatives up to the second degree. After this filtering, the Malagasy dataset consisted of 233 individuals and 579742 SNPs, and the global dataset consisted of 1178 individuals and 498926 SNPs.

SNPs in linkage were then removed by LD clumping (Privé et al. 2018), using tidypopgen. Those with an r2 value greater than 0.2 in a sliding window of 100 kilobases were removed. After pruning, the Malagasy dataset consisted of 223963 SNPs and the global dataset consisted of 188396 SNPs.

Principal components analysis was used to visualise Malagasy individuals compared to global comparative data. Non-malagasy populations were randomly subsampled to a maximum of 30 individuals to avoid bias through uneven sample sizes. This left a total of 336 individuals.

Unsupervised clustering in fastMIXTURE (Santander, Refoyo Martinez, and Meisner 2024) was run on the full, cleaned dataset of 1178 individuals and 498926 SNPs for 9 values of K (2to 10) and 3 runs per K. Subsequently, we analysed the output of fastmixture using the package tidygenclust (ref), which implements the clumppling algorithm of (Liu, Kopelman, and Rosenberg 2024) in R. We found a single mode for each value of K across 3 replicates.

ALDER (version 1.03) (Loh et al., 2013) was used to date the timing of admixture across different Malagasy groups, with the –mindis parameter set to 0.005. To relate estimated generation times to calendar years, we assumed a generation time of 29 years (refs).

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